MATH INQUIRERS PLACEMENT CHECK

Ages 12-13

For children after the PSLE • About 30 minutes • No calculator

Name: _____ Date: _____

How it works. Your child answers on their own. Read a question aloud if they ask, but do not help with the maths. If they are stuck, say "Try your best, or skip it", and move on. Write the answer in the box. Working on scrap paper is fine.

Warm-up

1 What is 2 1/2 - 3/4? Give your answer as a fraction or mixed number.

2 A shirt costs $60. It is discounted by 25%. What is the new price in dollars?

3 Two angles of a triangle are 52 and 63 degrees. What is the third angle, in degrees?

4 A car travels 135 km in 1 hour 30 minutes. What is its average speed in km per hour?

Main questions

5 A number machine doubles the input and then adds 5. The output is 31. What was the input?

6 How many prime numbers are there between 30 and 50?

7 The temperature is -4 degrees C at night. It rises by 9 degrees by noon, then falls by 12 degrees by evening. What is the evening temperature in degrees C?

8 A bag has 4 red, 3 blue and 5 green marbles. One marble is picked at random. What is the probability it is NOT green? Give a fraction in simplest form.

9 The mean of 5 numbers is 12. Four of the numbers are 8, 10, 15 and 11. What is the fifth number?

10 It is 10 o'clock now on a 12-hour clock. What time will it be 17 hours later? Give the hour only.

Brain stretchers

11 A robot faces North. It repeats this instruction 6 times: 'turn 90 degrees clockwise'. Which direction does it face at the end?

12 Using the digits 1, 2, 3 and 4, with no digit used more than once, how many different 3-digit even numbers can be made?

FOR THE TUTOR OR PARENT

Answer key and scoring

Accept equivalent forms (for example $2.05 and 2.05). One mark per question.

Warm-up (4 marks)

#	Question	Answer	Watch for
1	What is 2 1/2 - 3/4? Give your answer as a fraction or mixed number.	**1 3/4 (or 7/4)**	Check regrouping or conversion to improper fractions (10/4 - 3/4). Common error: 1 1/4 or 2 1/4.
2	A shirt costs $60. It is discounted by 25%. What is the new price in dollars?	**45**	Check 25% of 60 = 15, then subtract. Common error: answering 15.
3	Two angles of a triangle are 52 and 63 degrees. What is the third angle, in degrees?	**65**	Angle sum is 180. Common error: using 360 or arithmetic slip.
4	A car travels 135 km in 1 hour 30 minutes. What is its average speed in km per hour?	**90**	Child must convert 1 h 30 min to 1.5 h. Common error: 135 / 1.30 or 135 / 30.

Main questions (6 marks)

#	Question	Answer	Watch for
5	A number machine doubles the input and then adds 5. The output is 31. What was the input?	**13**	Look for working backwards (31 - 5, then halve). Common error: halving first or giving 26.
6	How many prime numbers are there between 30 and 50?	**5**	Primes are 31, 37, 41, 43, 47. Common errors: including 39 or 49, or counting 1.
7	The temperature is -4 degrees C at night. It rises by 9 degrees by noon, then falls by 12 degrees by evening. What is the evening temperature in degrees C?	**-7**	Check number line use: -4 + 9 = 5, 5 - 12 = -7. Common error: answering 7.
8	A bag has 4 red, 3 blue and 5 green marbles. One marble is picked at random. What is the probability it is NOT green? Give a fraction in simplest form.	**7/12**	Total is 12, not green is 7. Common error: 5/12, or 7/5 (comparing to green).
9	The mean of 5 numbers is 12. Four of the numbers are 8, 10, 15 and 11. What is the fifth number?	**16**	Total must be 60; the four add to 44. Common error: finding the mean of the four numbers.
10	It is 10 o'clock now on a 12-hour clock. What time will it be 17 hours later? Give the hour only.	**3**	10 + 17 = 27, 27 - 24 = 3 (or 17 mod 12 = 5, then 10 + 5 = 15 -> 3). Common error: answering 27 or 5.

Brain stretchers (2 marks)

#	Question	Answer	Watch for
11	A robot faces North. It repeats this instruction 6 times: 'turn 90 degrees clockwise'. Which direction does it face at the end?	**South**	6 x 90 = 540 = 360 + 180, so it ends facing the opposite way. Common error: East or West from miscounting turns; look for the child listing each turn.
12	Using the digits 1, 2, 3 and 4, with no digit used more than once, how many different 3-digit even numbers can be made?	**12**	Last digit 2 or 4 (2 ways), then 3 x 2 for the others. Common error: 24 (ignoring evenness) or listing incompletely.

YOUR CHILD'S RESULT

Which level fits?

Add up the marks, then find your row. This is a guide. You make the final choice, and you can change level free of charge before lesson 1.

Warm-up marks: _____ / 4 **Main + stretch marks:** _____ / 8

Main + stretch	What it suggests
6 to 8	Join **Ages 12-13 (P6)**. Your child has the foundations and will be well challenged.
4 or 5	Join **Ages 12-13 (P6)**, and expect some support. If the warm-up score is 2 or less, Ages 11-12 (P5) may be the more comfortable start.
0 to 3	We suggest **Ages 11-12 (P5)**. The ideas build step by step, and a confident start matters more than a fast one.

Strong?
Choose the level that fits the score, not the school year. A child who finds this easy will enjoy the challenge questions in class.

Not yet?
That is useful to know, not a failure. Starting one level down means more success and more joy in class.

Skills covered

Number machines and simple algebra; Primes; Negative numbers; Probability; Mean; Clock arithmetic; Counting possibilities; Turns and sequencing logic

Next step. Send the score and your preferred level to us when you enrol. Prices are the same at every level, and your tutor can talk it through first. Small groups of up to 8, live online.